

Efe Yılmaz

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SUMMARY

Backend-focused developer and Digital Game Design graduate, currently pursuing a non-thesis M.Sc. in Management Information Systems. Internship experience building REST APIs in C# / ASP.NET Core and a Unity mobile game; strong grounding in layered architecture, EF Core, PostgreSQL, Redis, JWT authentication, Docker, and automated testing with CI. Built a real-time matchmaking service load-tested with 1,000 concurrent players. Seeking a junior backend developer role.

EDUCATION

Dokuz Eylül University — Master's Degree, Management Information Systems (Non-Thesis) Online · 2026 – 2027 (Expected)

Istinye University — Bachelor's Degree, Digital Game Design · GPA 3.21 / 4.00 Istanbul · 2022 – 2026

EXPERIENCE

Software Developer Intern · *KPlus Bilişim Teknolojileri A.Ş. / Asteri Game Studio* Istanbul · Jun 2025 – Oct 2025

- Developed and tested backend features for business and application workflows using C#, ASP.NET Core, REST APIs, and SQL database operations.
- Built **Merge Vacuum**, a mobile casual puzzle game in Unity (C#, URP) for KPlus's in-house game studio, implementing the full gameplay loop and game-state UI across a 100+ script codebase.

PROJECTS

MatchForge — Real-Time Matchmaking & Lobby Service — *ASP.NET Core, SignalR, Redis, PostgreSQL, nginx, k6, Docker, xUnit · github.com/Efekoo/MatchForge*

- Built a real-time matchmaking & lobby service closing the full player loop: JWT auth with refresh-token rotation, skill-based matchmaking with expanding Elo/MMR windows, SignalR lobbies, and hand-rolled Elo rating persisted to PostgreSQL.
- Solved race conditions (duplicate matching, concurrent moves) with Redis distributed locks; implemented disconnect grace periods with reconnect and win-by-forfeit handling.
- Load-tested with k6: **1,000 concurrent players matched, p95 match time 2.0 s, 0% failed requests** across 2 nginx load-balanced replicas with a SignalR Redis backplane; Testcontainers integration tests on GitHub Actions CI.

Game Services API — *ASP.NET Core Web API, C#, EF Core, PostgreSQL, SignalR, Docker, xUnit · github.com/Efekoo/Game_Services_API · **live demo***

- Designed a four-layer solution (API / Business / Core / DataAccess) exposing seven domain controllers: authentication, user management, currency, inventory, market economy, friends, and leaderboards.
- Implemented JWT authentication with BCrypt password hashing; used Entity Framework Core against PostgreSQL for all data access, with Swagger documentation for every endpoint.
- Built a SignalR real-time leaderboard broadcasting live standings; containerized with Docker Compose and covered by 24 xUnit tests on a GitHub Actions CI pipeline.

Investor — Algorithmic Trading Bot — *Python, ccxt, pandas, SQLite, Optuna, Docker, pytest · github.com/Efekoo/Investor*

- Built a modular trading bot: exchange abstraction over ccxt, eight pluggable strategies, and a risk module covering position sizing, stop-loss / trailing stop, and a daily circuit breaker.
- Developed a backtest engine simulating fees, slippage, and partial fills with Optuna optimization; designed a layered live-trading safety model (paper / live modes, kill switch); 19 pytest tests in GitHub Actions CI.

AeroRecover — *Python, LangGraph, Google Gemini, Streamlit, Folium · github.com/Efekoo/AeroRecover*

- Multi-agent AI application generating post-flight recovery plans for pilots; six specialized agents orchestrated in a LangGraph workflow with safety filtering, delivered as a Streamlit app.

TECHNICAL SKILLS

- Backend:** C#, ASP.NET Core, Entity Framework Core, REST APIs, PostgreSQL, Redis, SQL Server, SignalR, JWT Authentication, BCrypt, Swagger / OpenAPI, AutoMapper, FluentValidation
- Testing & DevOps:** xUnit, Testcontainers, pytest, k6 Load Testing, Docker, Docker Compose, nginx, GitHub Actions CI
- Python:** pandas, ccxt, Optuna, LangGraph, Google Gemini API, Streamlit
- Game Development:** Unity, C#, Universal Render Pipeline
- Tools:** Git

LANGUAGES & CERTIFICATIONS

Turkish (Native) · English (B2)

Foundational C# with Microsoft — freeCodeCamp (Jul 2026) · **C# Intermediate** — Sololearn (Jan 2025) · **Python Intermediate** — Sololearn (Jan 2025)